FOOD INVENTORY MANAGEMENT SYSTEM

Micro Project Report

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By

NANDANA MOHAN PRN25CS087
MALAVIKA S PRN25CS074
JITHENDU S PRN25CS064
PRADEESHA S PRASAD PRN25CS096

Guided by
ANISH A AZIZ , ASSISTANT PROFESSOR , COLLEGE OF ENGINEERING PERUMON

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING PERUMON
CAPE - GOVT. OF KERALA, KOLLAM(D), KERALA

Table of Contents

Chapter No. Title

1. **Introduction**
2. **Scope of the project**
3. **Technologies used**
4. **Problem statement**
5. **Literature review or background**
6. **Implementation**
7. **Results and Discussion**
8. **Conclusion**
9. **References**

Introduction

The Food InventorySystemis a Java-based desktop application designed to simplify food inventory management. It allows users to add, update, search, and delete food items, manage stock quantities, organize products into categories, and track expiry dates. The system also identifies low-stock items and products nearing expiry, helping reduce food wastage and prevent stock shortages.

The application uses Java Swing to provide a graphical user interface and local file storage to maintain inventory records between sessions. It includes a dashboard for displaying inventory statistics and a reporting module for viewing important information. Overall, the project demonstrates the practical use of object-oriented programming, GUI development, data handling, and exceptionhandling in Java.

Scope of the project

TheFood Inventory System is designed to simplify the management of food items and maintain accurate inventory records. It allows users to add, update, search, and delete food items, manage categories, and adjust stock quantities. The system also helps monitor low-stock items and track expired or soon-to-expire products, reducing food wastage and preventing stock shortages.

The application provides a dashboard to display important inventory statistics and a reporting module to view inventory details. Using Java Swing, it offers a graphical user interface, while local file storage helps preserve records between sessions. Input validation and exception handling improve the reliability of the system.

The system can be used in small shops, food stores, and other settings that require basic food inventory management. In the future, it can be enhanced with database integration, barcode scanning, automated notifications, and multi-user access to support more advanced inventory operations.

Technology used

1.Java:Used asthe mainprogramminglanguage to implement the application's logic and inventory management operations.

2. Java Swing: Used to develop the graphical user interface (GUI), including login screens, forms, tables, and the dashboard.

3. Object-Oriented Programming (OOP): Used to organize the application into classes and objects, making the code modular and easier to maintain.

4. File Handling: Used to store and retrieve inventory data locally between application sessions.

5. VS Code:Used as the integrated development environment (IDE) for writing, running, and debugging the Java application.

6. Git and GitHub: Used for version control, source code management, and collaboration among team members.

Problem statement

Managing foodinventory manually can be time-consuming and may lead to errors in recording item details, tracking stock quantities, and monitoring expiry dates. Without a proper inventory management system, it becomes difficult to identify low-stock items, locate products, and maintain accurate records. These issues can result in stock shortages, food wastage, and inefficient inventory control.

The Food Inventory System aims to address these problems by providing a simple Java-based application to manage food items efficiently. It allows users to add, update, search, and delete inventory records, manage stock quantities, organize items by category, and monitor expired or soon-to-expire products. The system also provides inventory summaries and reports to support better decision-making. By reducing dependence on manual record keeping, the application aims to improve accuracy, save time, and simplify food inventory management.

Literature Review / Background

Traditional inventory managementsystems often rely on manual registersor spreadsheets to maintain stock records. These methods can be time-consuming and prone to errors, making it difficult to track quantities, monitor expiry dates, and identify low-stock items.

Modern inventory management applications improve efficiency by automating item recording, stock updates, searching, and report generation. Features such as expiry tracking and low-stock monitoring help reduce food wastage and prevent stock shortages.

Java provides object-oriented programming features for developing modular and maintainable applications, while Java Swing enables the creation of graphical user interfaces. Local file handling allows inventory data to be stored and retrieved without requiring a separate database system.

The Food Inventory System applies these technologies to provide a simple solution for managing food items, organizing categories, monitoring stock, tracking expiry dates, and generating inventory reports. It offers a practical approach to basic inventory management while demonstrating the application of Java programming and object-oriented concepts.

Implementation

MODULES OF THE FOOD INVENTORY SYSTEM

The Food Inventory System consists of several modules that work together to manage food items, monitor stock levels, track expiry dates, and maintain inventory records efficiently

1. Login Module

This module provides secure access to the application through a login interface. It validates the username and password and allows access to the dashboard when the entered credentials are correct.

2. Food Item Management Module

This module allows users to add, update, view, and delete food items. Each record contains details such as food ID, food name, category, quantity, unit, price, and expiry date. It also validates the entered data to maintain accurate inventory records.

3. Inventory Management and Search Module

This module displays all available food items in a tabular format. Users can search for items by food ID, name, or category, view their details, and select records for modification or deletion.

4. Category Management Module

This module organizes food items into categories, making inventory records easier to manage and locate. Users can add new categories and manage existing ones.

5. Stock Management Module

This module allows users to increase or decrease the quantity of food items based on stock received or consumed. It helps maintain accurate quantity records and prevents stock from being reduced below the available quantity.

6. Expiry and Low-Stock Monitoring Module

This module identifies expired food items, items approaching their expiry dates, and items with low stock levels. It helps users take timely action to reduce food wastage and avoid stock shortages.

7. Reports Module

This module provides inventory summaries and detailed reports, including total food items, low-stock items, expired items, and items expiring soon. It helps users understand the current inventory status.

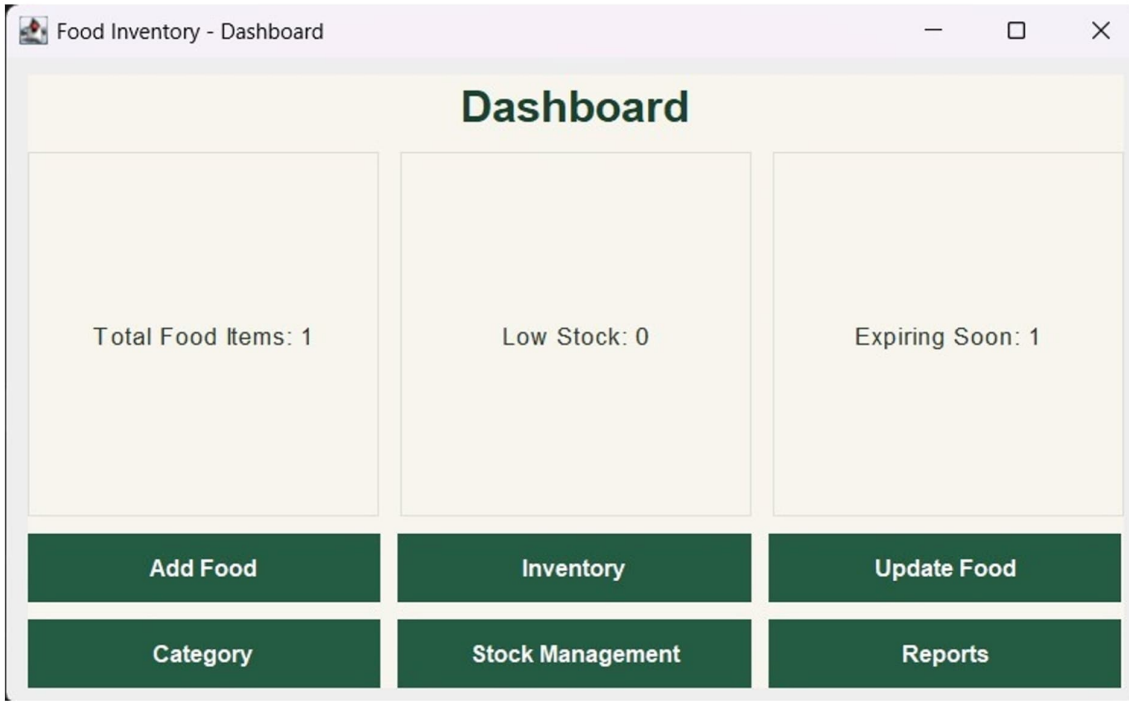
8. Data Storage and Validation Module

This module manages the saving and retrieval of inventory data using local file storage. It validates user inputs, handles errors, and preserves inventory records between application sessions

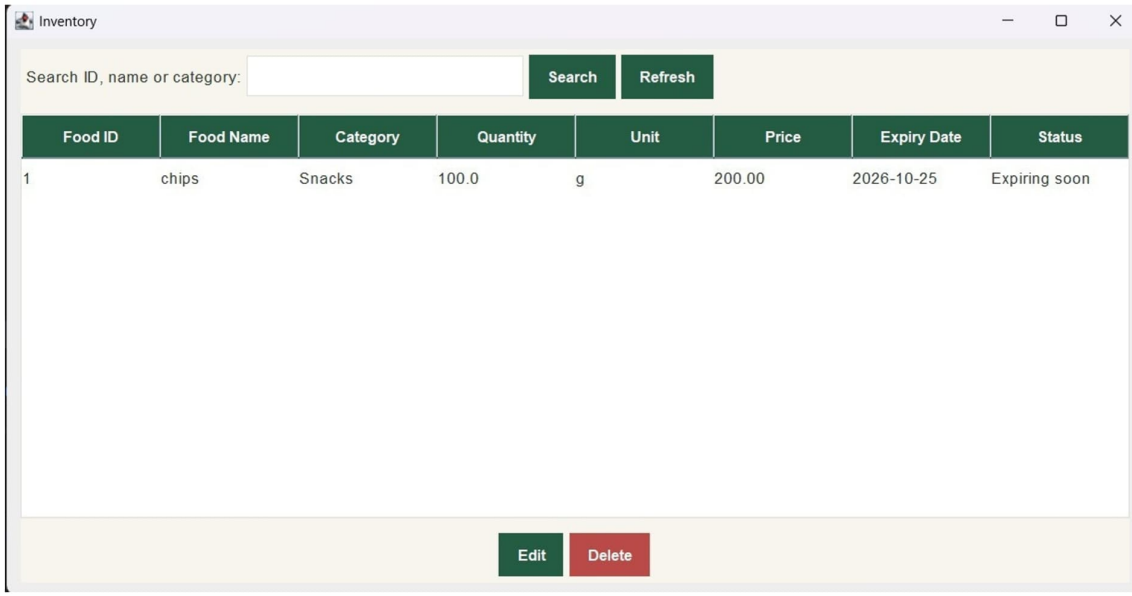
9. Dashboard Module

The dashboard serves as the main interface of the application. It displays important inventory statistics and provides navigation to the different modules, allowing users to access the system's features conveniently.

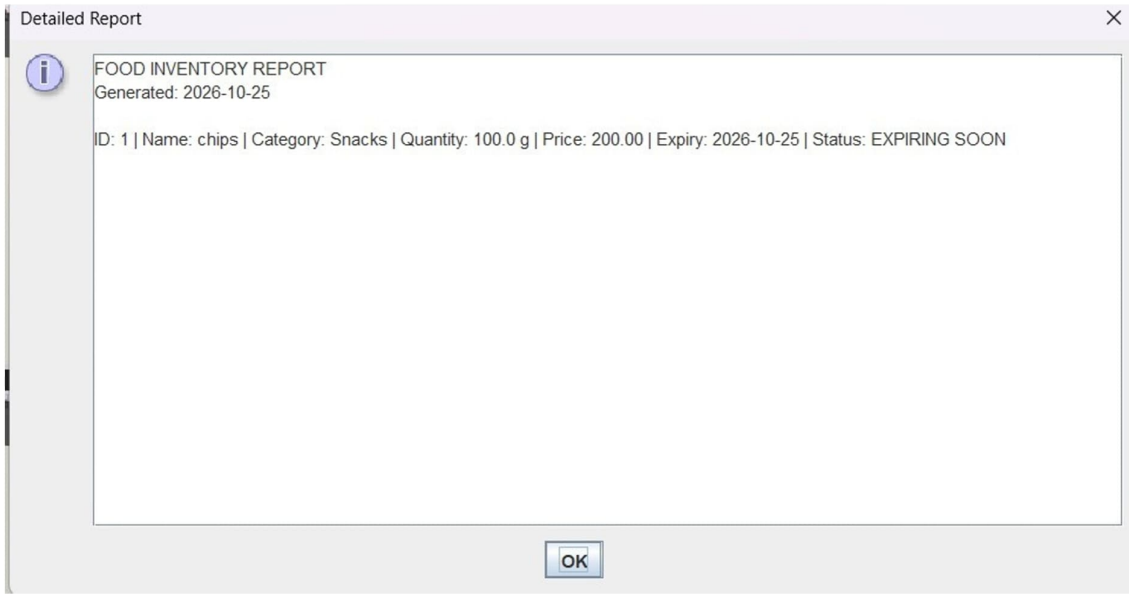
Dashboard Frame



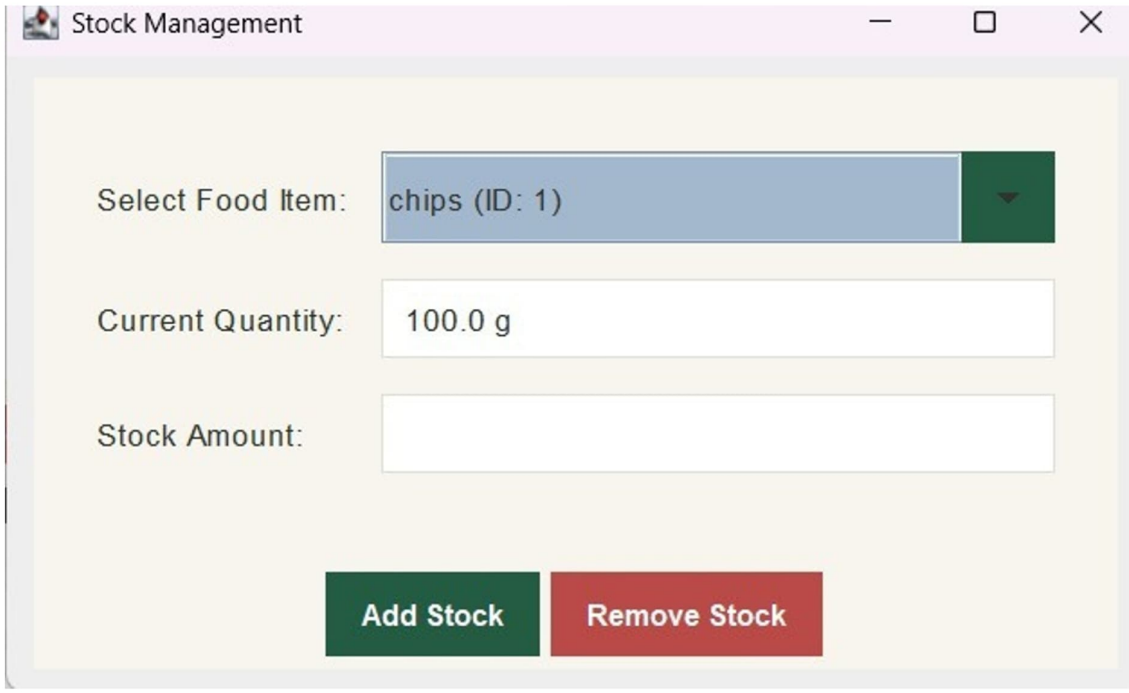
Inventory Frame



Food Inventory Report Frame

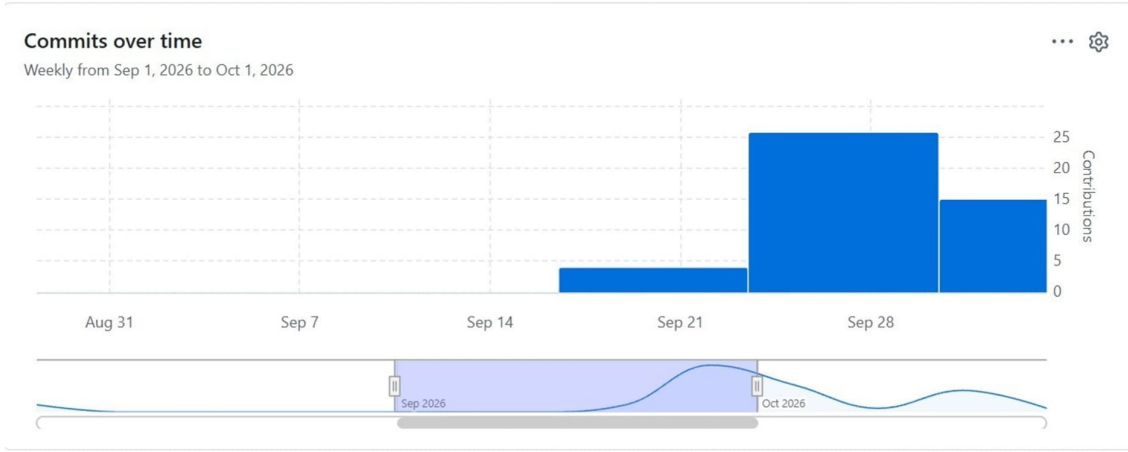


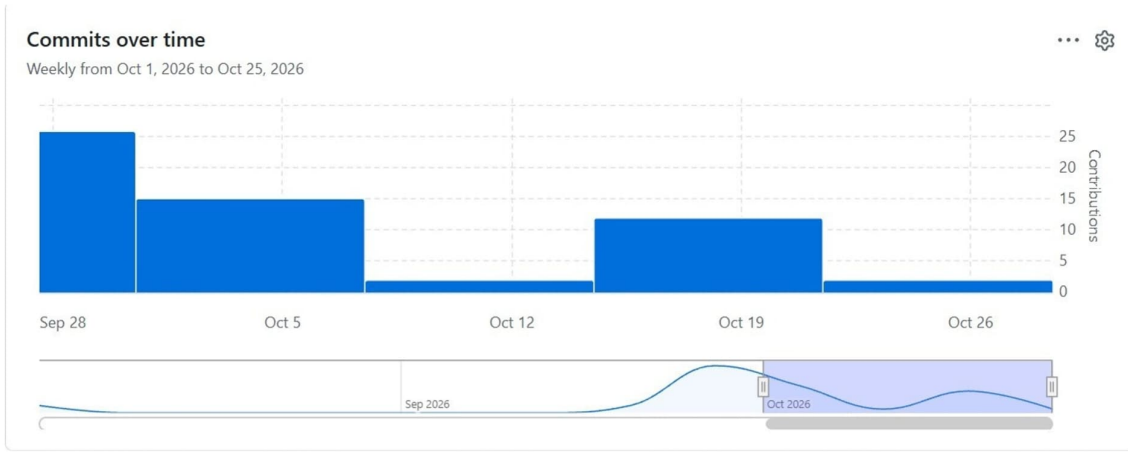
Stock Management Frame



Results and Discussion

TheFood Inventory System was successfully developed using Java and Java Swing. It allows users to manage food items, update stock, organize categories, and track expiry dates and low-stock items. The dashboard and reports provide useful inventory information, while local file storage maintains records between sessions. Overall, the system simplifies inventory management, reduces manual effort, and demonstrates the practical application of object-oriented programming and file handling in Java.





Conclusion

TheFood Inventory System provides an efficient way to manage food items, monitor stock levels, track expiry dates, and maintain inventory records. Features such as item management, low-stock alerts, and reports reduce manual effort and improve record-keeping accuracy. The application uses Java Swing for its graphical interface and local file storage to preserve data.

The project demonstrates key OOP concepts, including encapsulation, which combines data and methods within classes, and abstraction, which simplifies inventory operations. Inheritance and polymorphism are applicable where implemented in the code. Classes and objects organize the system, while exception handling helps manage errors and invalid inputs.

Overall, the project provides practical experience in Java programming, GUI development, file handling, and object-oriented design.

GitHub path

https://github.com/nandanamohan-tech/OOP/tree/main/src/FoodInventorySystem

Reference

1. Oracle, Java Documentation. https://docs.oracle.com/en/java/

2. Oracle, The Java Swing Tutorial. https://docs.oracle.com/javase/tutorial/uiswing/

3. Oracle, Java SE API Documentation. https://docs.oracle.com/en/java/javase/25/docs/api/

4. GitHub, Food Inventory System – Project Source Code. https://github.com/nandanamohan-tech/OOP/tree/main/src/FoodInventorySystem